RETAIL BUSINESS INSIGHTS

SQL Data Analysis Project

FERNANDO HASIHOLAN SINAGA

INTRODUCTION

Analisis data penjualan retail menggunakan MySQL

Mengeksplorasi pola penjualan, performa produk, dan perilaku pelanggan dari data mentah hingga menghasilkan insight yang actionable

Menjawab pertanyaan bisnis melalui query SQL seperti joins, aggregations, dan subqueries

"SALES PERFORMANCE BY BRANDS"

Key Question:

Brand mana yang menghasilkan pendapatan terbesar bagi perusahaan?

Goals:

- Identifikasi brand dengan pendapatan tertinggi
- Analisis volume penjualan untuk perencanaan inventory
- Prioritaskan kemitraan dengan brand potensial

Key Findings:

- **Trek** paling untung dengan revenue $129K dengan harga jual tinggi per unit
- **Electra** paling laris dengan 1,729 items terjual tapi harga produk lebih murah
- Bottom 3 brands lemah - total cuma kontribusi $13K

```sql
SELECT
    b.brand_name,
    COUNT(oi.item_id) AS total_items_sold,
    SUM(oi.quantity) AS total_quantity,
    ROUND(SUM(oi.quantity * oi.list_price), 2) AS total_revenue
FROM a.brands b
JOIN a.products p ON b.brand_id = p.brand_id
JOIN a.order_items oi ON p.product_id = oi.product_id
GROUP BY b.brand_name
ORDER BY total_revenue DESC;
```

brand_name	total_items_sold	total_quantity	total_revenue
Trek	1235	1839	5129381.61
Electra	1729	2612	1344143.79
Surly	605	908	1063135.82
Sun Bicycles	485	731	381919.69
Haro	220	331	207096.69
Heller	97	138	193798.71
Pure Cycles	258	376	166164
Ritchey	77	118	88498.82
Strider	16	25	4849.75

"TOP CUSTOMER IDENTIFICATION"

Key Question:

Siapa pelanggan paling menguntungkan berdasarkan riwayat pembelian?

Goals:

- Identifikasi pelanggan VIP
- Lihat distribusi geografis pelanggan top spender

Key Findings:

- **Pamela Newman** dari NY adalah pelanggan terbaik dengan total spending $37,801 dari 11 orders
- **New York** dominasi top 10 customers. 7 dari 10 pelanggan terbesar berasal dari NY
- Range spending cukup besar. Pelanggan teratas ($37K) vs terendah di top 10 ($27K) berbeda $10K

```sql
SELECT
    c.first_name,
    c.last_name,
    c.city,
    c.state,
    COUNT(o.order_id) AS total_orders,
    ROUND(SUM(oi.quantity * oi.list_price), 2) AS total_spent
FROM a.customers c
JOIN a.orders o ON c.customer_id = o.customer_id
JOIN a.order_items oi ON o.order_id = oi.order_id
GROUP BY c.customer_id, c.first_name, c.last_name, c.city, c.state
ORDER BY total_spent DESC
LIMIT 10;
```

first_name	last_name	city	state	total_orders	total_spent
Pamelia	Newman	Monroe	NY	11	37801.84
Abby	Gamble	Amityville	NY	7	37500.89
Sharyn	Hopkins	Baldwinsville	NY	10	37138.86
Lyndsey	Bean	Fairport	NY	11	35857.86
Emmitt	Sanchez	New York	NY	12	34503.82
Melanie	Hayes	Liverpool	NY	7	34390.88
Debra	Burks	Orchard Park	NY	11	30645.87
Elinore	Aguilar	San Angelo	TX	10	29661.83
Corrina	Sawyer	Troy	NY	8	29214.89
Shena	Carter	Howard Beach	NY	4	27618.95

"BEST SELLING PRODUCT BY CATEGORY"

Key Question:

Apa saja produk terlaris di setiap kategori?

Goals:

- Identifikasi bestseller di setiap kategori produk
- Optimalkan product mix dan pengaturan rak display
- Panduan untuk keputusan pembelian stok

Key Findings:

- **Electra Girl's Hawaii** dominasi Children Bicycles. 2 varian terjual 154 dan 145 unit
- Brand **Electra** kuasai 80% top products Prioritaskan display dan stok brand ini
- Produk >100 unit perlu stok prioritas dan hindari kehabisan stock bestseller

```sql
SELECT
    c.category_name,
    p.product_name,
    b.brand_name,
    SUM(oi.quantity) AS units_sold,
    ROUND(SUM(oi.quantity * oi.list_price), 2) AS revenue
from a.categories c
Join a.products p ON c.category_id = p.category_id
JOIN a.brands b ON p.brand_id = b.brand_id
JOIN a.order_items oi ON p.product_id = oi.product_id
GROUP BY c. category_name, p.product_name, b.brand_name
order by c.category_name, units_sold DESC;
```

category_name	product_name	brand_name	units_sold	revenue
Children Bicycles	Electra Girl's Hawaii 1 (20-inch) - 2015/2016	Electra	154	46198.46
Children Bicycles	Electra Girl's Hawaii 1 (16-inch) - 2015/2016	Electra	145	39148.55
Children Bicycles	Electra Cruiser 1 (24-Inch) - 2016	Electra	139	37528.61
Children Bicycles	Electra Girl's Hawaii 1 16" - 2017"	Electra	47	14099.53
Children Bicycles	Electra Townie 7D (20-inch) - Boys' - 2017	Electra	40	13599.6
Children Bicycles	Trek Precaliber 24 (21-Speed) - Girls - 2017	Trek	39	13649.61
Children Bicycles	Haro Shredder 20 Girls - 2017	Haro	39	8189.61
Children Bicycles	Electra Townie 3i EQ (20-inch) - Boys' - 2017	Electra	39	19109.61

"LOW STOCK ALERT"

Key Question:

Produk mana yang stoknya menipis dan perlu restocking?

Goals:

- Cegah kehabisan stok dan kehilangan penjualan
- Optimalkan waktu reorder produk
- Identifikasi produk yang perlu restocking segera

Key Findings:

- **4 produk Electra** stoknya habis (0 unit). Perlu restock urgent untuk hindari lost sales
- Mayoritas produk low stock dari **brand Electra** dan **Heller** . Prioritaskan reorder kedua brand ini
- 6 produk hanya tersisa 1 unit dan perlu action cepat sebelum benar-benar habis

```sql
select
    p.product_name,
    b.brand_name,
    st.store_name,
    sk.quantity AS current_stock
from a.products p
join a.brands b ON p.brand_id = b.brand_id
join a.stocks sk ON p.product_id = sk.product_id
join a.stores st ON sk.store_id = st.store_id
where sk.quantity < 10
order by sk.quantity ASC;
```

product_name	brand_name	store_name	current_stock
Electra Amsterdam Royal 8i Ladies - 2018	Electra	Rowlett Bikes	0
Electra Townie Original 21D - 2018	Electra	Baldwin Bikes	0
Electra Townie Original 1 - 2018	Electra	Santa Cruz Bikes	0
Electra Townie Original 1 Ladies' - 2018	Electra	Rowlett Bikes	0
Heller Shagamaw Frame - 2016	Heller	Baldwin Bikes	1
Trek Remedy 29 Carbon Frameset - 2016	Trek	Baldwin Bikes	1
Electra Cruiser 1 (24-Inch) - 2016	Electra	Baldwin Bikes	1
Haro SR 1.1 - 2017	Haro	Santa Cruz Bikes	1
Haro SR 1.2 - 2017	Haro	Rowlett Bikes	1
Trek Emonda S 5 - 2017	Trek	Santa Cruz Bikes	1
Trek Madone 9.2 - 2017	Trek	Santa Cruz Bikes	1

"GEOGRAPHIC CUSTOMER ANALYSIS"

Key Question:

Kota mana yang memiliki pelanggan dan penjualan tertinggi?

Goals:

- Identifikasi pasar utama untuk ekspansi
- Lihat konsentrasi revenue berdasarkan geografis
- Data untuk strategi marketing regional

Key Findings:

- **Mount Vernon, NY** adalah pasar terbesar dengan **20 customers** dan revenue **$117K** dari 60 orders
- **New York** state dominasi top markets dengan 5 dari 6 kota teratas berasal dari NY
- Pasar NY sangat potensial untuk ekspansi karena customer base dan revenue terkonsentrasi di sana

```sql
select
    c.city,
    c.state,
    COUNT(distinct c.customer_id) AS total_customers,
    COUNT(o.order_id) AS total_orders,
    ROUND(SUM(oi.quantity * oi.list_price), 2) AS total_revenue
from a.customers c
LEFT join a.orders o ON c.customer_id = o.customer_id
left join a.order_items oi ON o.order_id = oi.order_id
group by c.city, c.state
HAVING total_customers > 1
ORDER BY total_revenue DESC;
```

city	state	total_customers	total_orders	total_revenue
Mount Vernon	NY	20	60	117010.21
Ballston Spa	NY	17	52	110065.34
San Angelo	TX	12	53	109729.26
Baldwinsville	NY	7	37	105893.45
Howard Beach	NY	12	36	104250.51
Orchard Park	NY	11	48	101189.27
Canyon Country	CA	12	44	96243.47
Monroe	NY	11	44	93938.34
Houston	TX	9	44	90449.47
Astoria	NY	12	32	89650.56

"STORE PERFORMANCE COMPARISON"

Key Question:

Bagaimana perbandingan performa penjualan antar lokasi toko?

Goals:

- Bandingkan revenue antar lokasi toko
- Lihat pola traffic pelanggan tiap toko
- Identifikasi toko underperforming yang perlu support

Key Findings:

- **Baldwin Bikes (NY)** adalah toko terbaik dengan revenue tertinggi dan **1093 orders**
- **Santa Cruz Bikes (CA)** performa menengah dengan 348 orders, jauh di bawah Baldwin
- **Rowlett Bikes (TX)** perlu support karena hanya 174 orders dan performa paling rendah

```sql
select
    st.store_name,
    st.city,
    st.state,
    COUNT(distinct o.order_id) AS total_orders,
    SUM(oi.quantity) AS total_items_sold,
    ROUND(SUM(oi.quantity * oi.list_price), 2) AS total_revenue
from a.stores st
join a.orders o ON st.store_id = o.store_id
join a.order_items oi ON o.order_id = oi.order_id
group by st.store_name, st.city, st.state
order by total_revenue DESC;
```

store_name	city	state	total_orders	total_items_sold	total_revenue
Baldwin Bikes	Baldwin	NY	1093	4779	5826242.21
Santa Cruz Bikes	Santa Cruz	CA	348	1516	1790145.91
Rowlett Bikes	Rowlett	TX	174	783	962600.76

THANK YOU

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